UE DS52 : Foundations of Data Science
Presentation

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UTBM

« All models are wrong, but some are useful »

Georges Box

Outline

Course Language

... will be in English

Why?

- Essentially *all* machine learning literature is in English.
- Knowing the proper *terminology* is essential!
- Good to improve your English skills!
- Questions and answers in emails/homework/exams may be answered in French (However, this is not encouraged...).

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Prerequisites

- **Basic** probability theory
- Mathematics: **basic** knowledge of linear algebra, derivation, integration
- Computer science: **basic** programming skills
- **Basic** knowledge of python programming

Everything is **Basic** :)

Evaluations

- Two projects/homeworks during the **Training** sessions (30 points)
- Two computer-based assessments (MCQs) during **Tutorial** sessions (20 points)
- Two hours "**Final**" exam (50 points)
- Pass mark: 40 points (over 100) **and** a Final greater than 15 (over 50)

Issues addressed

- Part 1: **Fundamentals** (6 sessions)
  - Statistical inference: Maximum likelihood
  - Regression: linear regression
  - Classification: LDA, QDA, Naïve Bayes, Logistic regression
  - Unsupervised learning: clustering, density estimation, dimension reduction
  - Theory: Empirical risk minimization theory, penalization methods and bias-variance trade-offs
  - Transversal: Examine and understand the (characteristics of the) data $\Rightarrow$ Transformations, Visualizations, Handling missing data,...
- Part 2: **Machine Learning** (4 sessions with M. KAS)
  - Machine Learning: Neural Networks (perceptron)
  - Deep Learning (CNN)
  - Transformers
- Part 3: **Advanced Modeling** (3 sessions)
  - Support Vector Machine (SVM)
  - Boosting: ADABoost, XGBoost
  - Ensemble method: Random forest
  - Gaussian Process Regression (GPR)

Background reading

- Standard background reading:
  - C.M. Bishop, *Pattern Recognition and Machine Learning* (2006), Springer
  - T. Hastie, R. Tibshirani, and J. Friedman, *The Elements of Statistical Learning* (2015), Springer Verlag, https://web.stanford.edu/~hastie/Papers/ESLII.pdf
  - K.P. Murphy, *Machine Learning: a Probabilistic Perspective* (2012), MIT Press
  - S. Rogers, M. Girolami, *A First Course in Machine Learning* (2016), CRC Press
- Mathematics for machine learning background:
  - Marc Peter Deisenroth, A Aldo Faisal, and Cheng Soon Ong, *Mathematics for Machine Learning* (2020), Cambridge University Press, https://mml-book.github.io/
- Other resources:
  - D. Barber, *Bayesian Reasoning and Machine Learning* (2012), Cambridge University Press, http://web4.cs.ucl.ac.uk/staff/D.Barber/textbook/090310.pdf
  - R.O. Duda, P.E. Hart, and D.G. Stork, *Pattern Classification* (2nd ed. 2001), Willey-Interscience

Outline

Definitions

Data Science

The science of **collecting, storing, processing, analyzing, visualizing, and interpreting data** to extract knowledge and support decision-making.

Statistical Learning

A framework for understanding data that combines **statistical inference** with computational tools. It focuses on **interpretability, uncertainty quantification, and understanding the underlying data-generating process**.

Statistical Learning = the set of tools for understanding data

— *Hastie, Tibshirani, Friedman (2009)*

Machine Learning

A subfield of artificial intelligence that gives computers the ability to **learn without being explicitly programmed**. It focuses on algorithms that improve automatically through experience, with emphasis on **prediction accuracy and generalization to new data**.

- **Statistics** provides the foundation: inference, uncertainty quantification, model interpretation
- **Computer Science** provides the tools: algorithms, scalable computation, software
- **Domain expertise** provides the context: what questions are worth asking?

Statistical Learning vs. Machine Learning

Statistical Learning vs Machine Learning

- Statistical Learning
  - **Focus**: Inference, understanding mechanisms
  - **Emphasis**: Model interpretation, uncertainty
  - **Approach**: Probabilistic models, assumptions
  - **Question**: *How* does X affect Y?
- Machine Learning
  - **Focus**: Prediction accuracy
  - **Emphasis**: Algorithmic performance
  - **Approach**: Black-box optimization
  - **Question**: *What* will Y be given X?

In this course, we use mainly **Statistical Learning** — but we borrow tools from ML when they serve our goals.

The Two Main Pillars of Statistical/Machine Learning

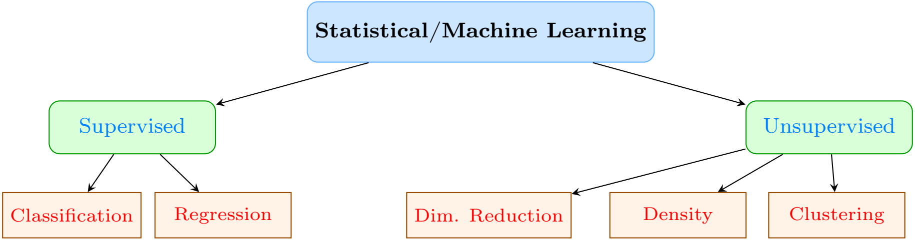

Figure: The main branches of Statistical/Machine Learning

Supervised Learning: Classification

Definition

We observe n pairs $(\mathbf{x}_i, y_i)$ where $y_i \in \{1, 2, \dots, K\}$ is a **class label**. Goal: predict y for new $\mathbf{x}$, and **understand which features discriminate between classes**.

Real Example: Email Spam Classification

- **Data**: 4601 emails, each described by:
  - Frequency of 57 words and characters: `george`, `free`, `!`, `$`, ...
- **Response**: `spam` (1) vs `email` (0)

	see	free	!	hp	your
`spam`	1.10	0.51	0.51	0.51	0.52
`email`	1.27	0.11	0.18	0.43	0.07
Difference	*-0.17*	*+0.40*	*+0.33*	*+0.08*	*+0.45*

Table: Average percentage of words/characters in spam vs. regular email

Statistical insight

Words like "`free`" and "`your`" show significant differences between classes. A statistical test (t-test) can quantify whether these differences are **real** or due to chance.
Next you can use embedding tools in order to build a classifier.

Supervised Learning: Regression

Definition

We observe n pairs (x_i, y_i) where $y_i \in \mathbb{R}$ is a **continuous response**. Goal: model $\mathbb{E}[Y|X = x]$ and understand the **relationship** between predictors and response.

Example: Prostate Cancer (Stamey et al., 1989)

- **Context**: 97 men about to undergo radical prostatectomy
- **Response**: log of Prostate Specific Antigen (`lpsa`)
- **Predictors**: clinical measurements

Variable	Description	Mean	Correlation with lpsa
lcavol	Log cancer volume	1.35	0.73
lweight	Log prostate weight	3.63	0.43
age	Age (years)	63.9	0.17
lbph	Log BPH amount	0.10	0.35
svi	Seminal vesicle invasion	0.22	0.37
lcp	Log capsular penetration	-1.39	0.44
gleason	Gleason score	6.97	0.28
pgg45	% Gleason 4 or 5	24.4	0.33

Question

Which variables show the **strongest marginal associations** with PSA?

Predictive Example: Same Data, Different Goal

What if we only care about prediction accuracy?

Method	**Test MSE**	**Interpretability**
Linear regression	0.40	High
Ridge regression	0.42	Medium
Lasso	0.43	High (sparse)
Random forest	0.38	Low
Gradient boosting	0.37	Low

Table: Prediction performance on prostate cancer data (10-fold CV)

Trade-off

The most accurate methods (boosting, random forest) are **black boxes**. Lasso gives us both good prediction **and** interpretability by selecting important variables.

Outline

One individual, many characteristics

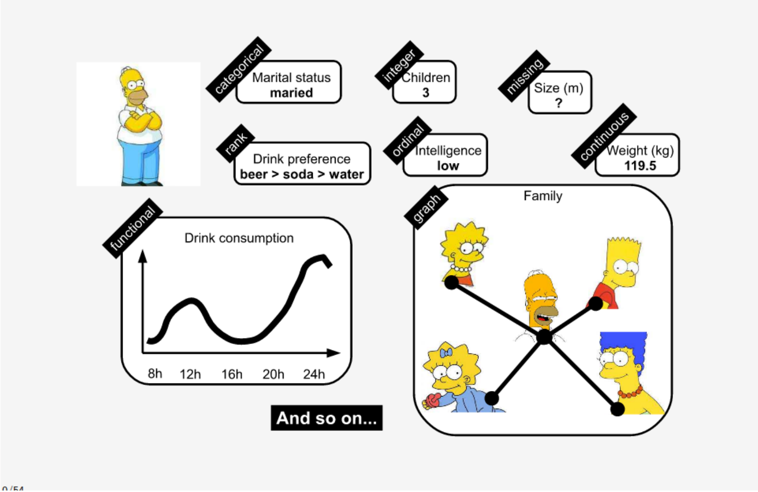

Figure: A representative heterogeneous datum about a representative individual

Tables

- Data refer to a *set* of observations ($\mathbf{x}_i$, $i = 1, \ldots, n$), with each **observation** (sometimes referred to as an **individual** or **entity**) groups together the values taken by the **variables** (or **features**, or **characteristics**) involved in the study.
- Sets of observations are generally represented by data **tables**, with each row corresponding to an observation (an individual, a datum) and each column (or group of columns, for a nominal variable) to a variable:

	$\mathbf{X}^1$	$\mathbf{X}^2$	$\ldots$	$\mathbf{X}^p$
$\mathbf{x}_1^\top$	x_{11}	x_{12}	$\ldots$	x_{1p}
$\mathbf{x}_2^\top$	x_{21}	x_{22}	$\ldots$	x_{2p}
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$\mathbf{x}_n^\top$	x_{n1}	x_{n2}	$\ldots$	x_{np}

Table: Standard table for a data set. Rows are referred as **Observations**. Columns are referred **Variables** (or **Series**, **Charateristics**, **Features**).

- n is the number of observations, p is the number of variables.

Coding categorical variables

One-Hot encoding

- For a categorical variable, its direct representation by a quantitative variable using ordinal coding is strongly discouraged

Teacher	Doctor	Technician
1	2	3

Table: Wrong way to encode categorical variables

- Such a representation introduces an order between modalities, an order that is **absent** from the meaning of the variable.
- For example, a prediction that is uncertain between `Teacher` and `Technician` will be assimilated to `Doctor`.
- The preferred representation for a categorical variable uses disjunctive (one-hot) coding

Teacher	Doctor	Technician
1|0|0	0|1|0	0|0|1

Table: Correct way to encode categorical variables

- The different modalities to be separated, but also has the disadvantage of increasing the size of the observations.

Coding categorical variables

One-Hot Encoding with Regression

Model income with a categorical predictor

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

where x_i is categorical with $L = 3$ modalities: Teacher, Doctor, Technician

Ordinal coding (WRONG):

$$x_i = \begin{cases} 1 & \text{if Teacher} \\ 2 & \text{if Doctor} \\ 3 & \text{if Technician} \end{cases}$$

The model becomes:

$$\begin{aligned} \text{Teacher: } & y = \beta_0 + \beta_1 \times 1 \\ \text{Doctor: } & y = \beta_0 + \beta_1 \times 2 \\ \text{Technician: } & y = \beta_0 + \beta_1 \times 3 \end{aligned}$$

⇒ Forces Doctor to be exactly midway between Teacher and Technician!

One-hot encoding (CORRECT):

$$(d_{i1}, d_{i2}) = \begin{cases} (1,0) & \text{Teacher} \\ (0,1) & \text{Doctor} \\ (0,0) & \text{Technician (reference)} \end{cases}$$

The model becomes:

$$y_i = \beta_0 + \beta_1 d_{i1} + \beta_2 d_{i2} + \varepsilon_i$$

$$\begin{aligned} \text{Teacher: } & y = \beta_0 + \beta_1 \\ \text{Doctor: } & y = \beta_0 + \beta_2 \\ \text{Technician: } & y = \beta_0 \end{aligned}$$

⇒ Each category gets its own offset!

Coding categorical variables

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Coding ordinal variables

thermometer coding

- Representing the values of qualitative variables is not as straightforward.
- Let's consider an ordinal variable, such as a LICKERT scale. Is it possible to represent its different values by numbers that respect the same order relationship, as in the following table?

Strongly disagree	Disagree	Neither disagree nor agree	Agree	Strongly agree
1	2	3	4	5

Table: Wrong way to encode ordinal variables

- This representation introduces identical distances between successive modalities. Is **disagreement** at equal distance from **strongly disagree** and **neither disagree nor agree**?
- An often-preferred representation is to associate as many binary numerical variables with this ordinal variable as there are different modalities, using the following binary coding:

Strongly disagree	Disagree	Neither disagree nor agree	Agree	Strongly agree
0|0|0|0|1	0|0|0|1|1	0|0|1|1|1	0|1|1|1|1	1|1|1|1|1

Table: Correct way to encode ordinal variables

- This representation preserves the order relationship between modalities, while avoiding imposing too strong constraints on the representation of successive modalities, since a subsequent modality is associated with the passage of a new binary variable to 1.

Coding ordinal variables

Thermometer coding

Ordinal variable example: Educational level

Level	1	2	3	4	5
Meaning	No diploma	High school	Bachelor's	Master's	PhD

Ordinal coding:

Level	Code
1	1
2	2
3	3
4	4
5	5

$\Rightarrow$ Assumes equal distance between levels

One-hot:

Level	Code
1	0000
2	1000
3	0100
4	0010
5	0001

$\Rightarrow$ Ignores order

Thermometer coding:

Level	Code
1	0000
2	1000
3	1100
4	1110
5	1111

$\Rightarrow$ Preserves order

Interpretation

Thermometer coding assumes that once you pass a threshold, you stay at that level:

$$y_i = \beta_0 + \beta_1 \mathbf{1}_{\text{level}\geq 2} + \beta_2 \mathbf{1}_{\text{level}\geq 3} + \beta_3 \mathbf{1}_{\text{level}\geq 4} + \beta_4 \mathbf{1}_{\text{level}\geq 5} + \varepsilon_i$$

Each β_j represents the **additional effect** of moving to level j or above.